# QIANQIAN(LEE) CUI

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## Education

**New York University**, New York, NY — May 2021 (Expected)
M.A. in Psychology
**Rutgers University**, New Brunswick, NJ — January 2019
B.A. in Psychology, Minor in Economics

## Honors and Awards

**Graduate Student Research Award,** New York University — 2020
**24th Annual MA Research Conference Best Poster, First Place Award,** New York University — 2020
**Threesis Academic Challenge Invited Contestant,** New York University (Canceled due to COVID-19) — 2020
**Dorothy and David Cooper Scholarship**, Rutgers University — 2017
**Dean's List,** Rutgers University — 2015- 2017

## Research Experience

**Masters Student Research Assistant** — September 2019 – Present
New York University, New York, NY
Social Neuroscience Lab (**PI: Dr. David M. Amodio**)
***Bias in AI Project***

- Main Hypothesis: AI facial recognition techniques would perpetuate the human bias in social categorization - hypodescent would be produced by the AI techniques when recognizing racially ambiguous faces, and the amount of hypodescent would depend on the racial composition of the training dataset
- Designed a computational approach examining potential racial bias in facial recognition techniques using Python (with PyTorch)
- Tested a popular convolutional neural network ResNet and fine-tuned the network with multiple public face datasets (unbalanced in terms of race and gender composition)
- Implemented a Siamese network that aims at comparing similarity of the two faces
- Generated 200+ morphed faces with WebMorph to test the network's performance
- Wrote python scripts able to only include the oval shaped facial areas within 2 minutes for 200+ faces
- Presented the preliminary findings of potential hypodescent in facial recognition techniques at NYU's 24th Annual Master's Psychology Research Conference and won first place award
- Wrote a research proposal and won Graduate Student Research Award

***Instrumental Reinforcement Learning in Social Decisions Project***

- Main Hypotheses: learned reward associations would be independent from subjects' declarative knowledge of the feedback probabilities in the probabilistic selection task; subjects would prioritize reward-based learning, rather than trait-based learning under special conditions such as economic scarcity
- Programmed probabilistic selection task examining the general process of value-based decision-making and attitude formation with JavaScript/HTML/CSS and psiTurk
- Programmed probabilistic selection task testing how people perceive traits in a scarcity condition with JavaScript/HTML/CSS and psiTurk
- Adopted the methodology of reinforcement learning
- Identified and fixed issues such as an ad blocker interfering with psiTurk online experiments

- Wrote code comments to make it easier for other lab members to read the scripts

**Masters Student Research Assistant** September 2019 - Present
New York University, New York, NY
Motivation Lab (**PIs: Dr. Gabriele Oettingen & Dr. Peter M. Gollwitze**r)

- Programmed and piloted surveys on belief and motivational processes in interpersonal interactions
    - Main hypothesis: subjects who believe they know things that are objectively unknowable would interpret ambiguous behaviors of others as more negative
- Generated animated images which are used in the Qualtrics surveys
- Fixed issues with Qualtrics having trouble hiding the next bottom while displaying multiple stimuli rapidly
- Coded response and performed content analysis
- Wrote Python scripts for data cleaning
- Attended and contributed to lab meetings

**Volunteer Research Assistant** February 2019 – August 2019
Nanjing University, Jiangsu, China
Reward, Competition and Social Neuroscience Lab (**PI: Dr. Yansong Li**)

- Designed, programmed and ran behavioral and eye-tracking studies examining how sex and visibility of sexual cues impact face and body perception among Chinese college students
    - Main Hypotheses: sex differences would be found in terms of subjective ratings (pleasantness, sexual arousal and attractiveness) and eye movements in the in face/chest/pelvic regions
- Assisted with stimulus design and data collection for EEG experiments examining how integration of primary/secondary reward cues influences executive functions
    - Main hypothesis: subject would optimize their performance in flanker tasks based on integrating incentive cues (none/juice reward/monetary reward/both rewards), and also display different neural activity dynamics based on incentive categories.
- Conducted experiments using E-Prime and Experiment Builder
- Cleaned and analyzed data in Python and R (~25k trials)
- Translated five questionnaires from English into Chinese
- Attended workshops on R and reinforcement learning
- Aided in building and maintaining the lab's new minimalist styled website with open source platforms (GitHub Pages & Hugo) link

**Undergraduate Research Assistant** January 2016 – May 2017
Rutgers University, New Brunswick, NJ
Close Relationships, Identity, and Stigma Lab (**Advisor: Dr. Diana T. Sanchez**)

- Collected data for multiple studies examining perception of fluidity of multiracial identity and attitude change
    - Main hypothesis: mono-racial subjects would perceive biracial persons who altered their identification based on the context to be less trustworthy and likable
- Interacted with confederates in front of real participants
- Marked attendance sheet and kept record of unexpected events

## Publications and Manuscripts

**Cui, Q**. & Li, Y. (in prep). Eastern Asian Erotic Stimuli Dataset: Sex Differences among Native Heterosexual East Asian Participants Influence Category-Specific Preferences toward Visual Erotic Stimuli from the Opposite Sex.

**Cui, Q**. & Li, Y. (in prep). Sex Differences in Eye Movements While Viewing Faces and Bodies Combined with Sexual Cues from the Opposite Sex.

## Posters and Presentations

**Cui, Q**., Berg, J. J., & Amodio, D. M. (2020, April). Hypodescent in Face Recognition Algorithms: Evidence and Effects of Training Dataset Racial Composition. Poster presented at the 24th Annual NYU MA Research Conference, New York, NY. (Best Poster, First Place Award)

**Cui, Q**. (2020, April). An Independent iOS App to Help with the Construction of Habits and Time Management Skills. Talk scheduled to be presented at NYU Graduate School of Arts and Science Threesis Academic Challenge (Qualifying Round), New York, NY. (Poster was selected to participate in the qualifying round; event was cancelled due to COVID-19)

## Course Project

**A Reinforcement Learning Approach for Understanding Grounded Natural Language Processing** 2020

- Completed a final group project for the doctoral-level course Computational Cognitive Modeling to explore language generalization and few shot learning in human cognition with a computational approach
- Collaborated with three other classmates with diverse backgrounds in Computer Science, Data Science and Engineering; all team members contributed equal amount of work
- Proposed and programmed a novel computational model by creatively combining the Seq2Seq model with a generative adversarial network and policy in reinforcement learning (inspired by the seqGAN model)
- Designed and collected data for an experiment testing language processing in human subjects
- Compared human subjects' results with the performance of our proposed model and wrote the final paper using LaTex

## Independent Project

**#17, the Habit Tracker** 2018 - Present

iOS habit tracker app with an outer-space theme link

- Created a fun app which aims to facilitate habit formation; users will play the role of an astronaut; they are encouraged to set up goals and record time they spend on those goals; stars will light up as a mental reward
- Independently developed graphic design, 3D models and scripts for the app
- Developed a low-poly art style with flat design UIs in order to generate a minimalist appearance
- Created animated 3D models of astronauts and planets with Blender
- Used particle systems in Blender to generate scenes of starry sky
- Completed graphic design with Photoshop
- Wrote Swift scripts which smoothly support UI design, data storage and animation display
- Improved product quality through code reviews and regular unit tests
- Currently working on a new version which will support multiple languages

## Additional Experience

**Part-Time Freelance Illustrator** 2018

link for artwork collection

- Conceptualized and created original graphics and illustrations from abstract ideas
- Designed and developed digital images with Clip Studio Paint and Adobe Photoshop

- Worked on projects including avatar and icon design based on clients' need
- Communicated efficiently with clients to make sure all the artwork meet their expectations

## Relevant Coursework

**Undergrad Level:**

- Social Psychology, Personality Psychology, Human Sexuality, Quantitative Methods in Psychology, Developmental Psychology, Abnormal Psychology, Abnormal Psychology Lab, Soul Beliefs, Cognition, Cognition Lab, Advanced Topics in Human Cognition

**Masters Level:**

- Psychology of Social Behavior, Cognitive Neuroscience, Decision Making, Cognitive Psychology, Masters Intermediate Statistics, Scientific Programming in Python & R, Research Methods & Experience

**Doctoral Level:**

- Prejudice and Stereotyping, Person Perception, Computational Cognitive Modeling

## Skills

**Coding Language**: Python, JavaScript, HTML/CSS, Swift
**Data Analysis:** Python, R, SPSS
**Experimental Programs**: psiTurk, E-Prime, Experiment Builder, PsychoPy, Qualtrics
**Image Processing:** Photoshop, Clip Studio Paint, Procreate
**3D Modeling**: Blender
**Other:** LaTex